



Coordinated Informative Path Planning for Multi-Robot Search in Open Fields

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Abstract

This study proposes an informative path planning algorithm for executing search missions cooperatively utilizing a multi-robot system, leading to a faster search over extensive areas such as oceans and large open fields. To facilitate efficient cooperative searching, we introduce a search area division algorithm based on the weighted Voronoi diagram and a coverage path planner inspired by the traveling salesman problem. Two key technical issues in cooperative search missions are efficient area coverage and accurate target localization, which involve a trade-off between exploration and exploitation. To address this trade-off issue, we present an algorithm that promotes active collaboration among the robots by leveraging the Fisher information matrix to optimize robot configurations and by employing a market-based strategy for effective compromise. The effectiveness of the proposed approach is demonstrated through simulation results.

Keywords Area partitioning · Cooperative search · Informative path planning · Multi-robot system

1 Introduction

Autonomous mobile robots can be utilized in a wide range of field applications due to their substantial advantages and capabilities in handling challenging tasks in areas with limited access, such as large open fields and ocean environments. These robots can conduct various search-related tasks, including search and rescue (SAR), environmental monitoring, and surveillance [1–5], with the aim of efficiently obtaining information distributed in the target area. When performing such tasks, it is crucial to balance two key aspects: exploration, which involves broadly surveying the entire given area to identify the presence and approximate location of target information, and exploitation, which entails carefully observing discovered objects of interest to reduce information uncertainty. This dual approach ensures

comprehensive coverage while allowing for detailed examination of significant findings.

Furthermore, when multiple autonomous mobile robots form a multi-robot system for cooperative search missions, information acquisition can be significantly accelerated, covering a wider area and greatly improving mission efficiency. However, simply deploying multiple robots can lead to unnecessary energy waste due to inefficient and redundant movements. To avoid this, cooperative movement among robots within a multi-robot system is crucial.

In this study, we propose an efficient search algorithm using a multi-robot system with the primary aim of maximizing the efficiency of the search task by enhancing cooperation among each robot in the multi-robot system. To achieve this, it is important for each robot to understand both its individual task progress and the overall progress of the multi-robot system, facilitating effective cooperation in the search mission. The proposed algorithm enables assigning suitable robots to exploration and exploitation tasks required for the search mission as the mission progresses and promotes effective cooperation between the robots.

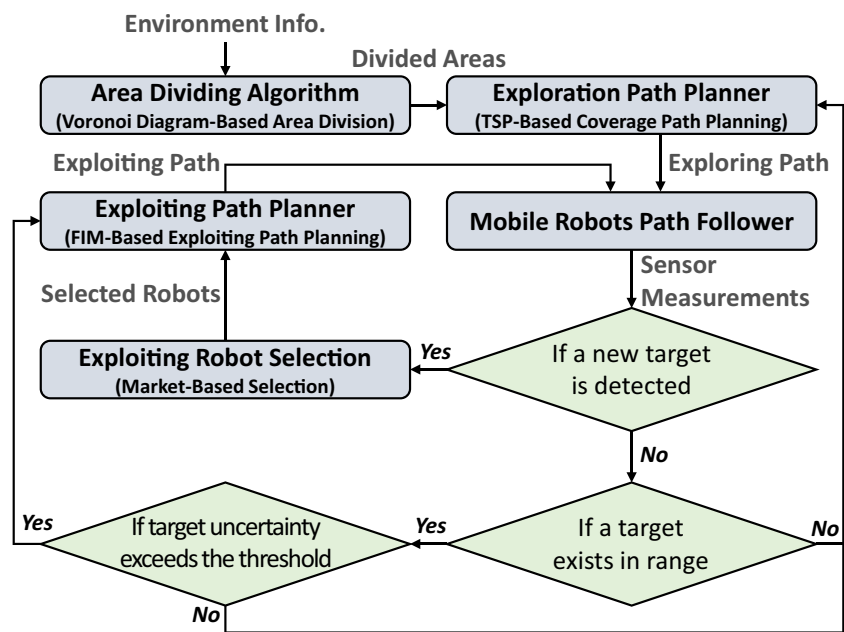
The schematic diagram in Fig. 1 illustrates the flow of the constituent algorithms and the data utilized in the methodology.

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Fig. 1 Flowchart of the path planning process for effective information acquisition in multi-robot systems. Detailed algorithms are represented in gray boxes, with information transmitted through each algorithm noted in gray text beside the arrows. Decision points in the algorithmic process are indicated by green diamond boxes



1.1 Related Work

Research on utilizing multi-robot systems for exploration, search operations, and other tasks is steadily expanding. Various attempts have been made to calculate the paths of multi-robot systems that can efficiently map a given area using an active SLAM approach [6–8], Frontier-based exploration algorithms [9, 10], Rapidly-exploring random tree-based methods [11, 12], and metaheuristic-based algorithms [13, 14]. To achieve effective exploration using multi-robot systems, approaches utilizing graph neural networks [15–17] or reinforcement learning [18–20] have been proposed to promote complete coverage. Additionally, learning-based planners leveraging distributed systems have also been introduced to address the coverage problem [21]. There have also been attempts to formulate the path planning problem for acquiring information in a given area using multi-robot systems in the framework of active information acquisition [22–25].

However, multi-robot systems encounter scalability issues due to their inherent computational complexity. To address this challenge, studies have been conducted on non-myopic scalable planning algorithms [26]. Additionally, algorithms for path planning in a distributed form have been developed to efficiently address communication issues [27–29], to design path planning methods based on effective information exchange [30], to represent large-scale exploration problems as graphs for facilitating distributed coordination [31], and to propose algorithms that resolve coupling constraints among multi-robot systems in distributed frameworks [32]. Meanwhile, research is being conducted to increase mission success rates in harsh environmental conditions [33–36].

While the importance of both exploration and exploitation in search missions is evident, most existing studies consider these two aspects separately by formulating and solving them as independent problems. This separation is due to the inherent trade-off between exploration and exploitation, making it challenging to achieve a balanced search strategy that simultaneously considers both elements.

To overcome this limitation, this study proposes a novel informative path planning algorithm for cooperative search in unknown areas using a multi-robot system. The proposed algorithm balances the use of an exploration path planner, which efficiently covers the entire area with a multi-robot system, and an exploitation path planner, which meticulously acquires information about targeted landmarks. This approach aims to address the limitations of previous research by integrating both exploration and exploitation strategies into a cohesive framework for more effective search missions.

1.2 Contributions

The potential contributions of this study can be summarized in the following:

- Distributing the search area among robots in a multi-robot system using the weighted Voronoi diagram concept, and calculating efficient exploration paths based on the Traveling Salesman Problem (TSP) that minimize the distance for each allocated area.
- Balancing exploration and exploitation strategies through the selection of market-based search algorithms, which

generate effective search paths while addressing their inherent trade-off.

- Conducting simulations across diverse search mission scenarios to demonstrate the superior performance of the proposed algorithm.

The remaining parts of this paper are structured as follows. Section 2 formalizes the problem this research aims to solve. The proposed algorithm is explained across Sections 3, 4, and 5. Section 3 addresses exploration search using a multi-robot system. Section 4 then describes the exploitation search based on the authors' earlier work [25]. Section 5 explains the methodology for resolving the trade-off relationship between the exploration and exploitation algorithms necessary for conducting the search mission, ultimately generating a balanced path. Section 6 presents performance evaluations through simulations. Finally, Section 7 summarizes and concludes the paper.

2 Problem Statement

In a scenario where the goal is to gather information in a specific area, consider a situation where multiple robots are deployed to search for and accurately localize multiple targets in unknown locations. Initially, it is assumed that no information about the search area is available, except for its boundaries, and that each robot has a finite sensing range. Consequently, exploration of the area is necessary to detect the presence of the targets and determine their approximate positions. However, the robots are assumed to have limited sensing capabilities, which necessitates dedicated observations for accurately estimating the target positions. Therefore, exploitation needs to be introduced to reduce the uncertainty associated with the target's location information.

Consider a situation where N robots form a multi-robot system, and there are an unknown number of target landmarks, assume to be M , that need to be located in the search area. Robots in the multi-robot system discover a landmark and acquire the location information in the form of the following measurement equation:

$$z_t = h(x_t) + v_t, \quad v_t \sim \mathcal{N}(0, \Sigma) \quad (1)$$

where z_t represents a measurement, x_t represents the state of a robot that has detected the target, $h(\cdot)$ represents the sensing function, and v_t represents white Gaussian noise with a covariance matrix of Σ .

Using Eq. 1, we can update the uncertainty associated with the information. In this case, the state of robot n at time step t is defined as $x_{n,t}$ with respect to the time horizon T . If we represent the trajectory of robot n as $\pi_n := x_{n,1}, \dots, x_{n,T}$, the entire path of the multi-robot system can be expressed

as $\pi_1, \dots, \pi_N \in \Pi$, and the problem can be formulated as follows:

$$\begin{aligned} & \underset{\Pi}{\operatorname{argmin}} \quad T_{fin}(\Pi) \\ & \text{subject to: } \max(p_1, \dots, p_M) < p_{goal} \\ & \quad CR(\Pi) = 1 \end{aligned} \quad (2)$$

where, $T_{fin}(\cdot)$ represents the time required to complete the entire mission, p_m represents the uncertainty of the target landmark's location after completing the mission, p_{goal} represents the target value for decreasing the uncertainty of the landmark's location, and $CR(\cdot)$ is the coverage rate of the searching area when performing the mission along the given path, taking a value between 0 and 1.

The proposed methodology aims to generate paths for a multi-robot system that minimize the time required to complete the entire mission. During this path generation process, it is necessary to ensure that the entire area is covered and the location uncertainty of all landmarks in the area is estimated to be below a target value.

3 Cooperative Exploration

In order to perform a search mission in an unknown region, it is important to efficiently visit unexplored areas, which is a challenge known as the coverage problem. The traditional strategy involves generating a lawn-mower type path, which can be an efficient method for exploring areas with a single robot without any prior map information [37, 38]. However, the optimal exploration path may vary significantly depending on the shape of the area and the configuration of the multi-robot system. Therefore, it is important to plan an efficient path considering the progress of the mission and the environment.

Furthermore, when conducting an exploration mission using multiple robots, it is crucial to distribute the area among the robots to prevent overlaps and ensure similar workloads. Considering these factors, this study formulates the coverage path planning problem as a traveling salesman problem (TSP) to dynamically generate exploration paths based on search situations.

3.1 TSP-based Coverage Path Planning

TSP is a problem that seeks to find the shortest path that visits each given waypoint once, which has been studied in various forms for many years. To adapt the coverage path planning problem into a TSP form, the generated path must cover the entire search area, and selecting waypoints is critical to avoid leaving any uncovered regions. Figure 2 illustrates the robot's coverage shape based on waypoints in a grid pattern.

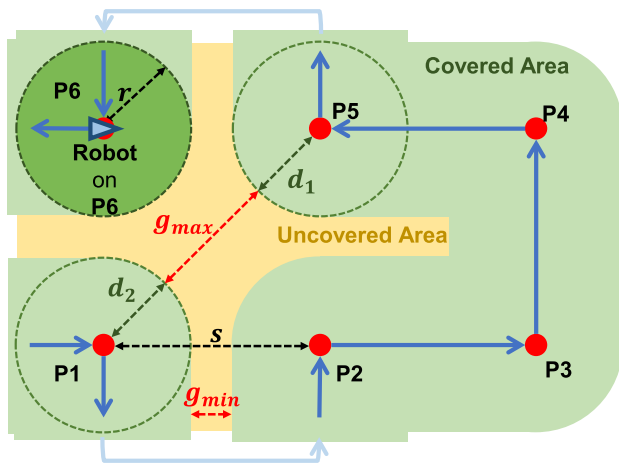


Fig. 2 The coverage level depends on the waypoint settings: To ensure there are no unvisited areas in the designated search area, it is essential to consider the sensing range of sensors and the path shape

The width of the uncovered area varies, with the narrowest, g_{min} , between two straight-line paths and the widest, g_{max} , between diagonally adjacent waypoints on the grid. Both can be expressed in terms of the circular sensing range, r , and the distance between waypoints, s .

$$g_{min} = s - 2r, \quad g_{max} = \sqrt{2}s - 2r. \quad (3)$$

If g_{max} is less than 0, the entire search area can be covered by visiting all specified waypoints. Therefore, the following equation must be satisfied when setting the waypoints.

$$s < \sqrt{2}r. \quad (4)$$

Additional considerations for the multi-robot aspects are required to apply the TSP to a multi-robot system. The simplest way is by utilizing multiple TSP (MTSP), which is a method that calculates an efficient route given the presence of multiple robots and waypoints [39]. However, MTSP has two main issues in our problem setting. Firstly, the computational complexity increases exponentially as the number of robots in the multi-robot system increases. Secondly, frequent path crossings may occur, causing overlapping between paths. This overlapping leads to redundant movements of each robot in area coverage. Therefore, this study proposes an efficient area division algorithm that considers the elements of the multi-robot system.

3.2 Voronoi Diagram-based Area Division

The Voronoi diagram divides an area based on multiple reference points, partitioning the space into regions that are closest to each respective point. In this study, we perform Voronoi cell decomposition using multiple robots as reference points

to allocate exploration areas to each robot. By segmenting the area with a Voronoi diagram and solving the TSP for each segment, we can mitigate the increase in computational complexity relative to the number of robots in a multi-robot system, ensuring it grows linearly. Additionally, this approach minimizes path crossing by assigning specific areas to each robot.

When there are N reference points for generating Voronoi diagrams, let's denote the i_{th} reference point as p_i . In this case, let set X represent a given region, and let d denote the distance between any two points in set X . Then, the Voronoi region R_j assigned to the j_{th} reference point can be expressed using the following equation [40]:

$$R_j = \{x \in X \mid d(x, p_j) \leq d(x, p_i) \text{ for all } i \neq j\}. \quad (5)$$

Depending on the form of the function d in Eq. 5, the shape of the Voronoi diagram can vary. When performing exploration, the workload burden on each robot can be considered proportional to the size of its assigned area. Consequently, one might consider using the Euclidean distance as the function d , given its direct relationship with the size of each cell generated by the Voronoi diagram. However, using solely the Euclidean distance may result in the allocation of uneven-sized areas to each robot. To overcome this limitation and utilize the Euclidean distance more flexibly, we employed the concept of a weighted Voronoi diagram as follows:

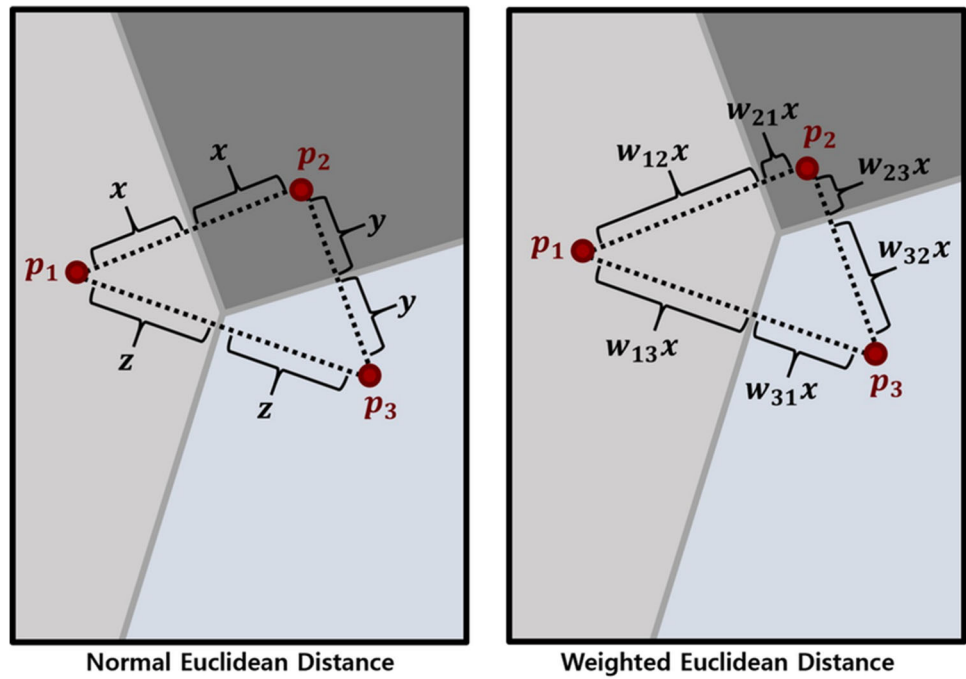
$$R_j = \{x \in X \mid w_{ji}d_e(x, p_j) \leq w_{ij}d_e(x, p_i) \text{ for all } i \neq j\} \quad (6)$$

where d_e denotes the Euclidean distance and matrix W represents the weights necessary for calculating the distance function. The element w_{ij} in matrix W is used as the weight for the distance calculation of p_i with respect to other coordinates when comparing the distances from p_i and p_j . Figure 3 illustrates how the definition of different Voronoi diagrams results in the partitioning of the regions. In a standard Voronoi diagram, boundary lines between regions are generated at points equidistant from the reference points. However, in a weighted Voronoi diagram, the positions of the boundary lines between partitioned regions are adjusted using weights by comparing the weighted distances from the reference points. When the Voronoi diagram is specifically defined, as in Eq. 6, accurate calculation of W is required to achieve a uniform division of the regions. In this study, we iteratively modify W by comparing the sizes of Voronoi regions.

Let A_i be the size of R_i , and define the relative difference as follows to compare the sizes of each region:

$$\phi_{ij} = \frac{A_i - A_j}{A_i + A_j}, \quad |\phi_{ij}| \in \Phi \text{ for all } i, j. \quad (7)$$

Fig. 3 Example of region partitioning by a Voronoi diagram. The size and the shape of the partitioned regions varies depending on the definition of the distance function



Using the threshold value η for the relative difference and the discount constant α , we can obtain $\mathbf{W}$ and uniformly divided area as follows:

Repeat until $\max(\Phi) < \eta$:

$$w_{ij} := w_{ij} - \alpha \phi_{ij} \text{ for all } i, j \quad (8)$$

In this equation, η was set to 0.01 to ensure convergence of the partitioned area size differences at approximately 2%. α , which influences the convergence speed and pattern of the weights, was set to 0.1 based on iterative simulations to determine an appropriate value.

4 Cooperative Exploitation

When there is a need to acquire specific information, minimizing uncertainty is a critical aspect of the search task. In [25], we introduced a path optimization algorithm designed to decrease the uncertainty associated with multiple information sources within the search area. This approach utilizes a multi-robot system equipped with sensors that cannot fully observe the environment with a single sensor due to their limited sensing capabilities (e.g., bearing-only or range-only sensing). To achieve comprehensive observability, it is crucial to enhance sensing efficiency by integrating measurements from each sensor. Considering the collaborative aspect of a multi-robot system, an exploitation path optimization algorithm is introduced with consideration of their spatial and temporal characteristics.

4.1 FIM-based Optimal System Configuration

The efficiency of information acquisition using a multi-robot system can vary depending on the location of each robot [41]. In other words, the configuration of the system and the positional relationship to the target information play a crucial role in information acquisition and affect the estimation quality based on the acquired information. The sensing effect according to the position of the multi-robot system can be observed in Fig. 4. If the system consists of robots using a bearing-only sensor and the robots are all arranged in a straight line (colinear) with the landmark, all these bearing measurements are identical, making it impossible to determine the landmark's location. On the other hand, if the robots are positioned at different angles to the landmark, the location of the landmark can be determined based on multiple independent angle measurements. Therefore, the configuration of the multi-robot system has a significant impact on information acquisition. The Fisher Information Matrix (FIM), which describes the impact of the configuration on the uncertainty of information, can be expressed as follows [42, 43]:

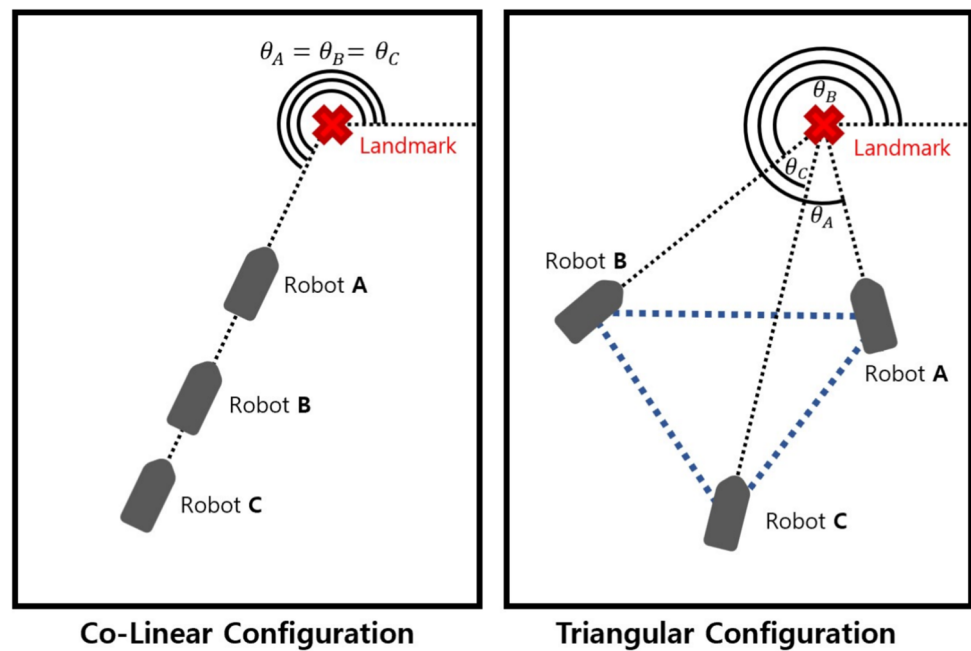
$$\mathcal{I} = \mathbf{H}_t^T \Sigma^{-1} \mathbf{H}_t \quad (9)$$

where $\mathbf{H}_t$ represents the Jacobian matrix for measurements, and Σ represents the noise matrix for measurements.

The uncertainty of the landmark's location information is related to the inverse of the determinant of the FIM:

$$B = \frac{1}{|\mathcal{I}|}. \quad (10)$$

Fig. 4 The effect of configuration on sensing performance in multi-robot systems. In a multi-robot system utilizing bearing sensors, if the robots are arranged in a linear configuration, it becomes difficult to expect the benefits of utilizing multiple sensors since only the same measurement can be obtained. On the other hand, by performing sensing in a distributed configuration of the constituent robots, it is possible to enhance sensing performance by acquiring multiple measurements



As depicted in Fig. 5, when there are multiple landmarks, the uncertainty of each landmark is individually determined by the relative configuration it forms with the multi-robot system.

Finally, the overall uncertainty level caused by the current configuration of the multi-robot system for all M landmarks can be expressed as $B_{\text{total}} = \sum_{m=1}^M B_m$. Since the overall uncertainty level is a function of the position of each robot, by taking partial derivatives, we can determine in which direction each robot should move to efficiently reduce the uncertainty of landmarks in the next time step:

$$u_n = - \begin{bmatrix} \frac{\partial B_{\text{total}}}{\partial x_n} \\ \frac{\partial B_{\text{total}}}{\partial y_n} \end{bmatrix} \quad (11)$$

where u_n represents the control input of robot n .

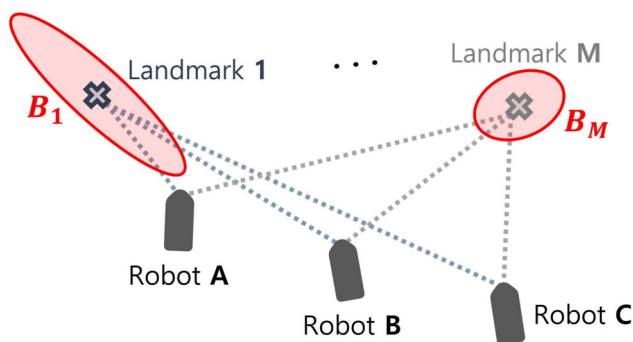


Fig. 5 Illustration of multiple landmarks and their individual uncertainties based on the relative configuration with the multi-robot system

4.2 Path Refinement

The impact of the multi-robot system's configuration on the uncertainty of landmarks can be examined based on the FIM, with a particular focus on their spatial characteristics. However, focusing solely on spatial characteristics leads to myopic planning, where robots tend to approach the closest landmarks, resulting in inefficient path selection. To address this issue, we propose a methodology that not only considers spatial characteristics but also incorporates temporal characteristics, aiming to predict reductions in information uncertainty while taking into account computational constraints within the multi-robot system. We use an Extended Kalman Filter (EKF) to update landmark information and compute the covariance matrix, which can be compressed by assuming that the covariance matrix of the prediction step is an identity matrix multiplied by a scalar.

$$\begin{aligned} P_t &= \Gamma P_{t-1} \\ &= (I - \sigma H_t^T (\sigma H_t H_t^T + R)^{-1} H) P_{t-1}. \end{aligned} \quad (12)$$

Equation 12 shows that the covariance matrix changes over time by a factor of Γ . Using the Γ matrix, it is possible to predict how the uncertainty ellipse will change in one time step, which is referred to as the predicted convergence rate (PCR) and defined as follows:

$$\text{PCR} = \sqrt{\lambda_{\Gamma_1} \lambda_{\Gamma_2}} = \gamma \quad (13)$$

where λ_{Γ_1} and λ_{Γ_2} are the eigenvalues of Γ .

The number of time steps κ required to reach the target uncertainty can be determined:

$$p_{goal} = \gamma^\kappa p_0, \quad \kappa = \frac{\log \frac{p_{goal}}{p_0}}{\log \gamma} \quad (14)$$

where p_0 is the landmark uncertainty level and p_{goal} is its desired goal level.

Therefore, landmarks with high κ should be considered more important when performing path planning, and κ can be used as a weight for Eq. 11 to calculate the refined control input to the multi-robot system as follows:

$$u_n = - \sum_{m=1}^M \kappa_m \begin{bmatrix} \frac{\partial B_m}{\partial x_n} \\ \frac{\partial B_m}{\partial y_n} \end{bmatrix}. \quad (15)$$

5 Exploration and Exploitation Trade-Off

The balance between exploration and exploitation has persistently posed a challenge across diverse fields. This study addresses this trade-off, with exploration involving the acquisition of information by visiting unexplored areas and exploitation entailing the thorough examination of gathered information. Conflicts may arise from choosing between these two search approaches. In this study, we employ the framework of a market-based approach to enable each robot to make appropriate choices between exploration and exploitation [44, 45].

5.1 Market-based Search Algorithm Selection

For the search mission, all robots start exploring an unknown region without any prior information. Once desired information is collected during exploration, the process to select robots for further exploitation of this information begins, which is referred to as **auction**. The auction involves the participation of all possible combinations of robots in the multi-robot system.

In the case of a multi-robot system with N robots, ranging from μ_1 to μ_N , the following set relationships can be defined:

$$\mu_1, \dots, \mu_N \in \mathbf{M}, \quad \mathcal{P}(\mathbf{M}) = \{\emptyset, \{\mu_1\}, \{\mu_2\}, \dots, \{\mu_1, \dots, \mu_N\}\}. \quad (16)$$

Here, $\mathcal{P}(\mathbf{M})$ represents the power set of the set $\mathbf{M}$. Each element of $\mathcal{P}(\mathbf{M})$ can be considered as a set composed of robot combinations participating in the auction. Considering a set with N elements, the size of the power set is 2^N . Therefore, in a multi-robot system consisting of N robots, including the empty set, there can be 2^N robot combinations participating in the auction. An example scenario is depicted

in Fig. 6. When multiple robots exploit the same information, they determine the exploitation path for acquiring information according to the path planning strategy explained in Section 4. Meanwhile, robots that are not selected in the auction continue to perform exploration in the remaining areas. The exploiting robots will resume exploration once they have reached the target uncertainty value of the target information.

5.2 Auction Bidding Structure

When an auction is held to select the robot to perform the exploitation of certain information, all participating robots in the auction will bid. The goal of the proposed algorithm is to complete the search task in the minimum amount of time. Hence, each combination of robots bids based on the estimated time required to exploit the information. It takes into account the time required to reach the information and the time required to update the uncertainty of the information through the measurement of each robot.

In the auction process, robots from the multi-robot system participate in teams as represented in Eq. 16. Let us assume that a team attempting to bid consists of n robots in total. We can calculate the arrival time t_r it takes for the sensing range of each robot to start including target information. The time can be sorted in ascending order as $t_{r1} < \dots < t_{rn}$ to arrange and index the robots. With each arrival time of information, the number of robots observing the information increases by one. That is, between t_{ri} and t_{ri+1} , a total of i robots will observe the target together. In this context, through the PCR relationship equation of Eq. 14, the time interval t_{oi}

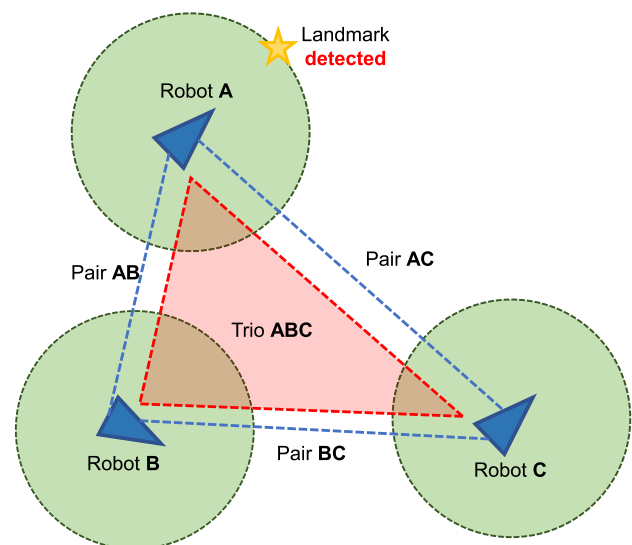


Fig. 6 To facilitate cooperative missions in a multi-robot system, a robot ensemble is chosen via market-based selection for exploitation tasks. In a multi-robot system, every possible combination of robots participates in the auction process

during which i robots observe the target can be represented. If the PCR value when observing information with i robots is denoted as γ_i , then t_{oi} can be expressed as follows:

$$t_{oi} = \begin{cases} t_{ri+1} - t_{ri} & 0 < i < n. \\ \frac{\log \frac{p_{goal}}{\bar{p}_{n-1}}}{\log \gamma_n} & i = n, \quad \bar{p}_{n-1} \geq p_{goal}. \\ 0 & i = n, \quad \bar{p}_{n-1} < p_{goal}. \end{cases} \quad (17)$$

$$\bar{p}_{n-1} = \left(\prod_{i=1}^{n-1} \gamma_i^{t_{oi}} \right) p_0 \quad (18)$$

As indicated in Eq. 17, when i ranges from 1 to $n - 1$, the calculation is based solely on the arrival times of each robot. However, when i equals n , the calculation should include consideration of how quickly target uncertainty can be reduced based on PCR. Meanwhile, when the $\bar{p}_{n-1} < p_{goal}$, t_{on} takes a negative value. This implies that uncertainty can reach the p_{goal} before the n_{th} robot arrives. Therefore, when there is no need for all n robots to observe the landmark, t_{on} is treated as 0. The relationship between t_{ri} and t_{oi} can be observed in Fig. 7. From this relationship, the final bidding function can be calculated as follows:

$$f_{cost}(S) = t_{rn} + t_{on} \quad (19)$$

where S is a set representing the robot combinations for cost calculation. As evident from the Eq. 19, the cost is based on the predicted time required for mission execution; therefore, a lower value is preferable. Consequently, the auction structure in this context favors the lowest bidder as the winner.

Finally, when the auction opens, all participating teams calculate their bidding costs in ascending order of the number of robots comprising each team. If a team can propose a lower

bidding cost than the current lowest bidder at the time of calculation, that team can assume the position of the lowest bidder. After all teams have attempted to bid, the final team remaining with the lowest bid is selected as the robot team to perform the exploitation task.

However, conflicts may arise due to equal bidding costs during this process. In such cases, the team currently occupying the lowest bidder position retains that status. In this auction, the order of bidding attempts is determined by the number of robots in each team, with teams having fewer robots bidding first. When teams have identical bidding costs, it is more efficient from a mission execution perspective to use fewer robots. Therefore, even if a team with the same cost as the lowest bidder appears later, it does not replace the current lowest bidder.

The pseudocode representing the flow of the entire algorithm is shown in Alg. 1.

Algorithm 1 Cooperative search with multi-robot systems.

Input: Robot $i = 1, \dots, n$, Search Area R ,
Weight Matrix W as all-ones matrix
Output: Search Path Π

```

1: while  $\max(\Phi) > \eta$  do  $\leftarrow$  Weighted Voronoi diagram iteration
2:    $R_n \leftarrow (6)$ 
3:    $\Phi \leftarrow (7)$ 
4:    $W \leftarrow (8)$ 
5: end while
6:  $\Pi \leftarrow$  TSP-based exploration path
7: while  $CR(\Pi) < 1$  do
8:   Explore along the search path
9:   if Landmark is discovered then
10:    Select robots  $k = 1, \dots, K$  with Eq. 19
11:    while  $p_{landmark} > p_{goal}$  do
12:       $\mathcal{I}, \gamma, \kappa \leftarrow (9)(13)(14)$ 
13:       $\Pi' \leftarrow$  FIM-based exploitation path  $\leftarrow (15)$ 
14:    end while
15:  end if
16: end while

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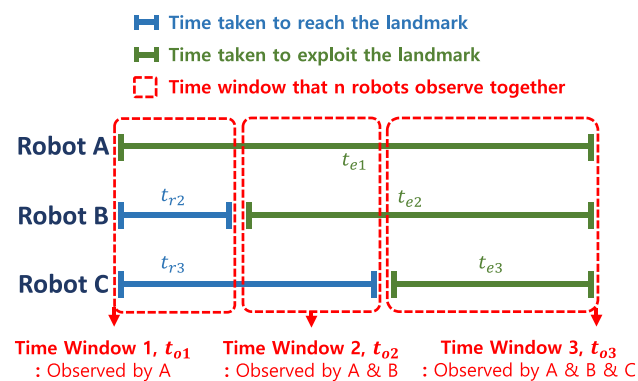


Fig. 7 The bidding cost calculation method for the combination of three robots. The time required for exploiting varies depending on the locations of each robot and target landmarks, as well as the sensing range

6 Performance Evaluation

In order to evaluate the performance of the proposed algorithm, simulations were conducted. The performance examined through the simulation includes: how quickly the search task is completed for a given search area, and how accurately the target landmark located at a random location in the search area is localized.

6.1 Simulation Setting

The simulation employs a multi-robot system to perform a search task in a given area with no prior information, aiming to localize the target landmarks within the entire

region. The multi-robot system consists of three robots, each equipped with a bearing-only sensor with a sensing range of 50m, enabling efficient information acquisition through collaborative measurement acquisition among the robots. The multi-robot system completes its search mission when it has discovered all targeted landmarks and achieved 100% coverage of the entire area.

The search area has a size of $700\text{m} \times 700\text{m}$ and contains four target landmarks. The deploying area is defined as a location 50m outside the search area, and the multi-robot system starts the search from this area. After setting a depot where the multi-robot system will stand by within the deploying area, the multi-robot system is deployed with a 100m gap based on the center position of the depot.

For the comparison group, we formulated the search mission as an MTSP and executed the mission based on the solution paths generated for a multi-robot system. The solutions for the MTSP were derived using two distinct approaches. The first approach employed a reinforcement learning-based method utilizing a transformer model [20], which has recently gained significant attention in combinatorial optimization tasks. Transformer architectures have been shown to model complex relationships between nodes in routing problems using self-attention mechanisms to generate scalable and computationally efficient solution paths. Following the procedure in [20], a pre-trained model for 50 nodes was fine-tuned for 15 hours to adapt it to the simulation scenario, resulting in a model capable of solving the MTSP problem for 100 nodes. The second approach utilized a powerful meta-heuristic method, the Genetic Algorithm (GA), which is widely recognized for its ability to approximate near-optimal solutions for NP-hard problems like MTSP [46, 47].

In addition, the lawnmower path-based search algorithm, which is still most commonly used for search missions across various domains [48–50], was also selected as the control group for performance comparison with the proposed

algorithm. The lawn-mower pattern coverage is a simple and effective method for ensuring complete coverage of the target area. However, since the proposed algorithm uses a multi-robot system, it is necessary to distribute the search task area for each robot before the path planning stage to use the lawn-mower pattern for coverage. To perform the given process, previous studies have utilized conventional Voronoi diagrams [51, 52]. However, in this study, we define and employ weighted Voronoi diagrams, as described in Subsection 3.2, to achieve more effective task execution. Accordingly, we aim to compare two different area distribution methods. Once the area division is completed, the swath parameter becomes important in the lawn-mower pattern. A wide swath allows for faster completion of the task but may not provide sufficient measurements of the target. In contrast, a narrow swath provides sufficient measurements but may result in delayed completion of the task. To compare the performance of the proposed algorithm in different situations, two comparison groups were set up: one with the swath as calculated by the proposed algorithm, and the other with a narrower swath. The names and characteristics of the described algorithms can be found in Table 1.

To evaluate the performance of the proposed algorithm, simulations were conducted using Monte Carlo simulation. A total of 100 Monte Carlo simulations were performed, with the location of the target landmark randomly changed for each simulation, and the location of the depot where the multi-robot system begins the search task was also randomly changed within the deploying area.

The simulations were conducted using MATLAB and the Optimization Toolbox, which provides robust solvers for mixed-integer linear programming (MILP). MATLAB was chosen for its reliability and flexibility in implementing such algorithms. The Optimization Toolbox was specifically utilized to solve the TSP within the coverage path planning framework, ensuring efficient and accurate optimization.

Table 1 Explanation of algorithms used in the simulation

Algorithm	Path Generation Type	Area Dividing Method	Swath Size
Proposed	TSP	Weighted VD	Calc. by Eq. 4
MTSP+TR	MTSP	—	Same as proposed
MTSP+GA	MTSP	—	Same as proposed
LM+WVD+NoS	Lawn-mower Pattern	Weighted VD	Same as proposed
LM+WVD+NaS	Lawn-mower Pattern	Weighted VD	20% less than proposed
LM+VD+NoS	Lawn-mower Pattern	Conventional VD	Same as proposed
LM+VD+NaS	Lawn-mower Pattern	Conventional VD	20% less than proposed

The control group utilized the Multiple-Traveling Salesman Problem (MTSP) solution path and a lawn-mowing pattern-based path. To obtain the MTSP solution, two independent approaches were employed: a reinforcement learning-based method using a transformer and a genetic algorithm (GA). Abbreviations used include "TR" for Transformer, "GA" for Genetic Algorithm, "MTSP" for Multiple-Traveling Salesman Problem, "LM" for Lawn-Mower, and "VD" for Voronoi Diagram. In the lawn-mower pattern, swath widths were classified as "NoS" (Nominal Swath), matching the width of the proposed algorithm, and "NaS" (Narrower Swath), which is 20% narrower than the nominal swath.

The mission in the simulation requires the robots to explore the given target area without gaps and accurately obtain the location information of landmarks within the area. Therefore, all paths are generated to achieve a coverage rate of 100%. To compare the mission efficiency of each algorithm under these conditions, two metrics are measured: mission completion time and the accuracy of the acquired landmark location information. Mission completion time is defined as the duration from when the multi-robot system departs from the depot, performs the search task based on the generated paths, and continues until all robots stop after fully covering the unexplored regions. Landmark location information is acquired using a bearing-only sensor, while the state is continuously updated through an EKF. The accuracy of the acquired landmark location information is determined at the mission completion time using the final covariance matrix updated by EKF. The accuracy is calculated based on the eigenvalues of this matrix, which represent the size of the uncertainty ellipse.

6.2 Simulation Results

The simulation was performed according to the described scenario settings, and a quantitative performance evaluation of the proposed algorithm was conducted through Monte Carlo simulations. The time taken for search task completion and the level of landmark location uncertainty after task completion are illustrated in Figs. 8 and 9, respectively. Figure 8 shows that the proposed algorithm completes the task the fastest. Initially, it was expected that the proposed algorithm would be slightly slower compared to the comparative algorithms which perform the task without other reactive path generation processes. However, the simulation

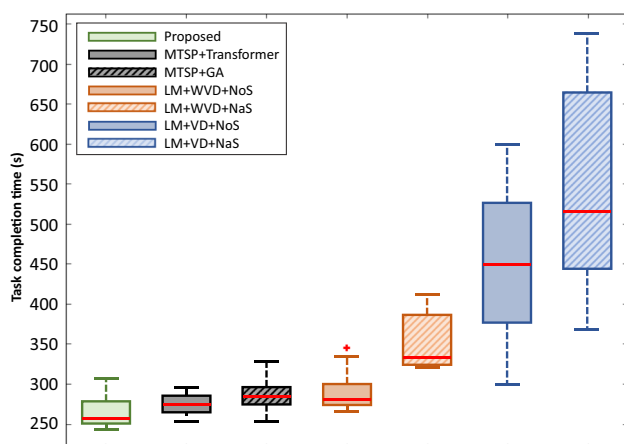


Fig. 8 Results of mission completion time comparison based on the path generation algorithm for information acquisition. The results are visualized as box plots, where the median is represented by a red line within the box. Outliers are marked with red crosses outside the box and whiskers

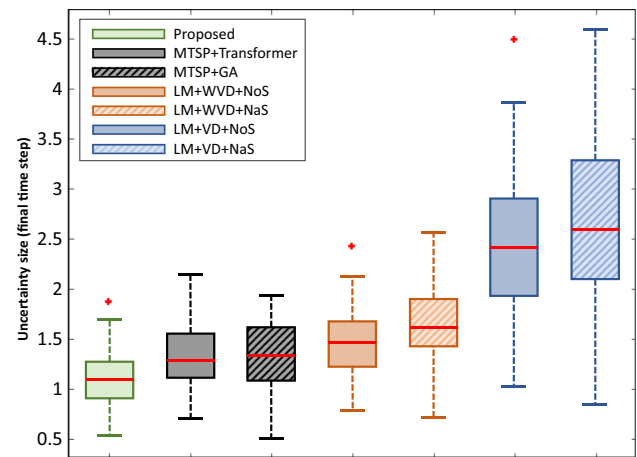


Fig. 9 Results of landmarks' position uncertainty comparison based on the path generation algorithm for information acquisition. The results are visualized as box plots, where the median is represented by a red line within the box. Outliers are marked with red crosses outside the box and whiskers

results contradicted this expectation, and this is likely due to the path generation characteristics of each algorithm.

For the MTSP solution paths, the transformer-based method achieved the fastest mission completion time among the comparison algorithms, second only to the proposed algorithm, with the smallest performance gap. However, due to the inherent complexity of MTSP, which is significantly higher than that of single-vehicle TSP, the transformer-based method exhibited slightly slower mission completion times compared to the proposed algorithm. Moreover, when deriving solutions using a transformer-based approach, obtaining a single solution requires a very short amount of time. However, significant computational effort is necessary to adapt the model to new problem scenarios whenever the number of nodes in the problem changes. Considering this, obtaining path-planning solutions through transformer-based learning cannot be described as achieving results with minimal computational cost solely due to its short execution time. On the other hand, for the GA-based MTSP solution paths, the GA continuously improves the solution quality starting from an initial guess, enabling the generation of search mission paths that complete the task relatively quickly, achieving the third-best performance after the proposed algorithm and the transformer-based solution paths. However, overlapping paths among the robots often occurred, leading some robots to follow redundant routes from a search mission perspective. This overlap frequently resulted in extended mission completion times compared to the proposed algorithm. Additionally, due to the high computational complexity of the MTSP, obtaining paths using the GA typically took about 20 minutes to converge to a final solution for an average problem instance. This highlights a significant difference

in computational efficiency compared to the proposed algorithm, which can generate final paths within a second.

In the case of a lawn-mower pattern path, repetitive movements in one direction occur for the given waypoints. Therefore, long diagonal movements may occur frequently, depending on the problem situation. On the other hand, the proposed algorithm calculates the shortest path for the given waypoints based on the TSP solution and moves accordingly, thereby showing a significant advantage in terms of time. Among the algorithms generating lawn-mower patterns, it can be seen that the task completion time for LM+WVD+NoS and LM+WVD+NaS, which utilize a weighted Voronoi diagram, is relatively short. This is because they decrease the overall task completion time by evenly distributing the areas among the robots. Additionally, LM+WVD+NaS and LM+VD+NaS, which use a narrower swath, take longer to complete the task compared to the other methods.

For the uncertainty reduction performance, Fig. 9 shows that the proposed algorithm is the most effective in reducing the uncertainty in landmark position compared to other algorithms. Unlike other algorithms, it reactively generated efficient paths for reducing uncertainty through robot cooperation in situations where exploitation was necessary. On the other hand, the MTSP approach demonstrated a notable reduction in uncertainty, second only to the proposed algorithm. MTSP+TR demonstrated a lower average uncertainty level compared to MTSP+GA. However, in certain cases, MTSP+GA exhibited an uncertainty level even lower than that of the proposed algorithm. This can be attributed to the overlapping paths among robots, as previously mentioned. When overlapping occurs near a target landmark, the excessive exploration around this landmark leads to a significant decrease in uncertainty. Consequently, while the MTSP+GA method effectively reduces uncertainty, it does so at the cost of mission efficiency due to redundant exploration. This over-exploitation, resulting from the redundant paths, ultimately lowers the operational efficiency in certain scenarios.

Comparing the results of other benchmark algorithms, it can be seen that the results of LM+WVD+NoS and LM+WVD+NaS are better than those of LM+VD+NoS and LM+VD+NaS because the algorithm took advantage of the weighted Voronoi diagram to greatly reduce the time for

landmark uncertainty propagation while gaining a temporal advantage. In addition, LM+WVD+NaS and LM+VD+NaS attempted to increase the reliability of the information obtained using a narrower swath. However, this led to a significant increase in mission time, thus extending the propagation time of landmark location uncertainty. Therefore, it can be concluded that simply narrowing the swath has no significant effect on reducing uncertainty.

The quantitative performance comparison results are summarized and shown in Table 2. t_f and p_f represent the mission completion time and the uncertainty of landmark positions at the mission endpoint, respectively. The average and standard deviation of these values measured in the simulation are presented. As shown in Table 2, the path planning based on the proposed algorithm demonstrated superior performance in terms of both mission completion time and information accuracy. This was followed by the MTSP-based path planners, with the lawn-mowing pattern path planners showing comparatively lower mission efficiency.

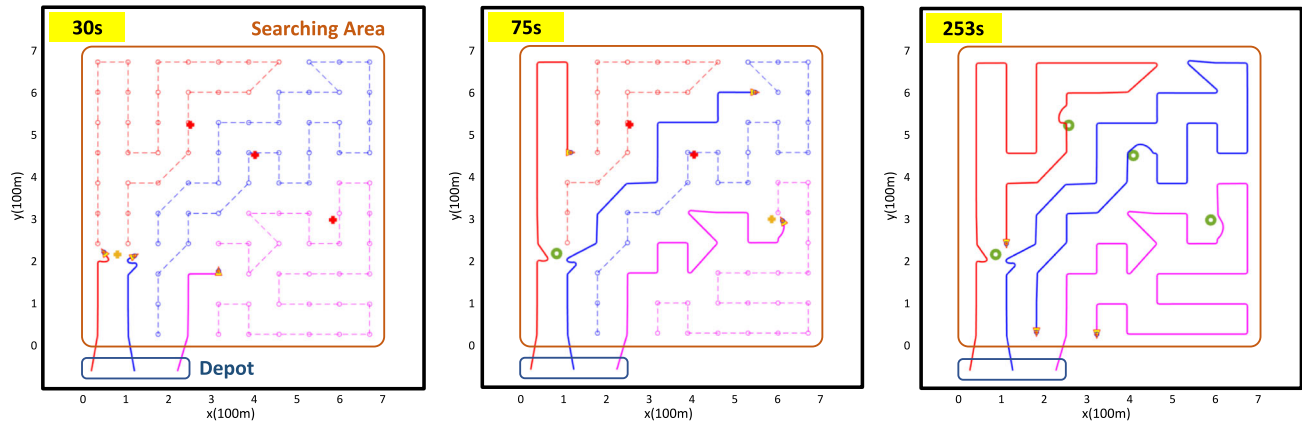
Path planning results generated by the proposed algorithm for three different scenarios are shown in Fig. 10, depicting each robot's actual maneuvering path over time along with the initial search paths. For the landmarks, those not yet discovered are marked with a red cross, landmarks targeted for exploitation to reduce uncertainty after discovery are marked with a yellow cross, and landmarks for which information acquisition has been completed with the desired accuracy are marked with a green circle. In this figure, Scenario 1 represents one of the path planning outcomes derived from a Monte Carlo simulation environment. The initial exploration path planned for search task is represented as the dotted lines, showing an equal distribution of areas based on a weighted Voronoi diagram and the shortest distance search path based on the TSP solution. In the first figure at 30 seconds, two robots deviate slightly from the planned path because one robot discovers a landmark and starts exploitation for uncertainty reduction. In this situation, the two robots near the landmark are selected as exploitation robots based on the market-based selection algorithm. They cooperate along the optimal configuration to efficiently reduce the uncertainty of the landmark's location. In the next figure at 75 seconds, another landmark is discovered and the two robots previously performing exploitation return to the exploration task for the

Table 2 The overall simulation results for performance evaluation of the proposed algorithm

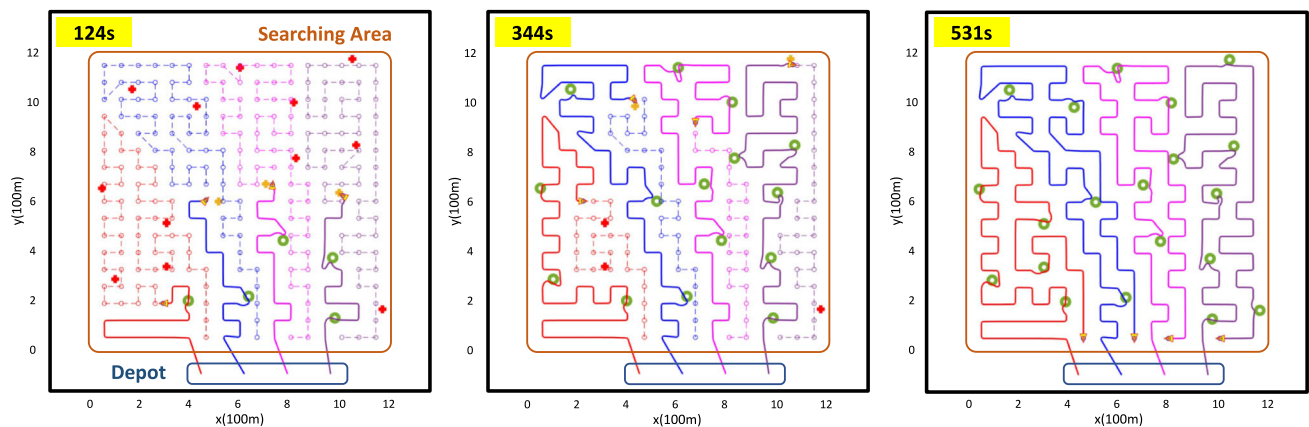
	Proposed	MTSP +TR	MTSP +GA	LM+WVD +NoS	LM+WVD +NaS	LM+VD +NoS	LM+VD +NaS
Avg(t_f)	263.42	274.22	285.10	291.84	352.48	464.35	553.75
Std(t_f)	16.78	11.48	15.25	25.99	32.26	93.93	120.79
Avg(p_f)	1.10	1.32	1.34	1.46	1.64	2.44	2.66
Std(p_f)	0.26	0.29	0.34	0.34	0.36	0.69	0.81

t_f represents mission completion time, and p_f denotes the uncertainty of discovered landmark positions

Scenario 1: 3 Agents / 4 Landmarks / 700m X 700m Rectangular Area



Scenario 2: 4 Agents / 20 Landmarks / 1,200m X 1,200m Rectangular Area



Scenario 3: 4 Agents / 15 Landmarks / 1,200m X 1,200m Non-rectangular Area

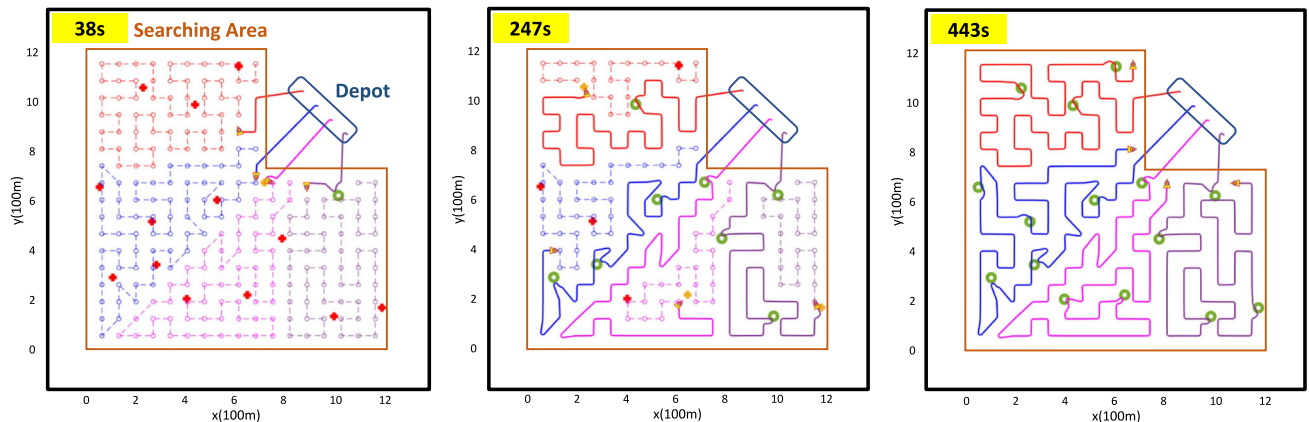


Fig. 10 Path planning results for three scenarios using the proposed algorithm. Each scenario differs in the number of robots comprising the multi-robot system, the number of landmarks present in the area, the area size, and the shape of the area, with specific scenario characteristics detailed above each result. Paths were generated for initial exploration, and as time passed, separate exploitation paths were created when the desired information was discovered, ensuring reliability for that specific information. The paths are visually distinguished as

follows: initial planned paths are represented by dashed lines, while the paths followed during actual mission execution are represented by solid lines. Furthermore, each robot's search path is assigned a distinct color to differentiate them. Additionally, landmarks are color-coded to indicate their status: landmarks not yet discovered are marked with a red cross, landmarks targeted for exploitation are marked with a yellow cross, and landmarks for which information acquisition has been completed with the desired accuracy are marked with a green circle

uncovered area. Finally, in the last figure at 253 seconds, the search task is completed by visiting all the waypoints, achieving a 100% coverage rate and sufficiently reducing the uncertainty of the landmark locations in the area.

In Scenarios 2 and 3, simulations were performed under more complex mission conditions. In Scenario 2, the simulation involved utilizing four multi-robots and acquiring information on a total of 20 landmarks over an area approximately three times larger. For this scenario, area partitioning was appropriately performed to ensure uniformity in the size of the search areas. Both the TSP-based coverage paths and the exploitation paths aimed at reducing information uncertainty were effectively utilized according to the mission conditions, enabling the acquisition of all landmark information in the area with the desired accuracy.

Scenario 3 also employed four multi-robots and maintained the same area size as Scenario 2, but changed the previously simulated rectangular area to a non-rectangular area. Despite the change in area shape, Voronoi-based area partitioning was effectively applied. The proposed path planning algorithm successfully acquired information on all 15 landmarks present in the area, as illustrated in the figure. This simulation scenario demonstrated that the proposed methodology has the potential to effectively utilize multi-robot systems even in search areas with more complex shapes, rather than just standard rectangular forms.

6.3 Computational Complexity Analysis

The proposed algorithm generates the initial global path through two main processes: constructing a weighted Voronoi diagram and generating a TSP-based coverage path for sub-areas. First, the weighted Voronoi diagram is constructed by iteratively adjusting weights to find the desired partitioning of the area. On a 2D plane, an efficient algorithm for computing Voronoi diagrams with n reference points has a computational complexity of $\mathcal{O}(n \log n)$ [53]. Since appropriate weight values are determined through a maximum iteration count ζ , the computational complexity of the weighted Voronoi diagram can be expressed as $\mathcal{O}(\zeta \cdot n \log n)$.

Next, after partitioning the area, a TSP-based coverage path is generated. In this study, the solution is extracted using integer programming. For ψ nodes, the initial problem setup includes $\mathcal{O}(\psi^2)$ variables and constraints, and solving this problem generally involves exponential computational complexity. When using a multi-robot system to cover an area with ψ nodes, the robots in the system are assigned ψ/n nodes each, where n is the number of robots. Therefore, while the TSP-based coverage path process inherently has exponential computational complexity, increasing the number of robots reduces the size of individual TSP problems and results in repeated computations proportional to the number of robots.

As such, the impact of robot count and environmental conditions on the computational complexity of the proposed algorithm is intertwined in a nonlinear manner due to various factors. To investigate how robot count and node count for coverage affect computation, numerical analysis based on simulations was conducted. Simulations were performed by increasing the number of robots from 3 to 10. As the exploration area grows, more nodes are required for coverage; thus, simulations were also conducted by increasing the node count from 50 to 150 in increments of 10. For each simulation, the time required for initial global path planning was measured, and 50 simulations were performed for each scenario.

Figure 11 illustrates the average computation time for each scenario. The x-axis represents the number of robots in the multi-robot system, while the y-axis shows the average computation time. For better visibility, only a subset of scenarios is displayed in the graph. As shown in Fig. 11, for cases with relatively fewer nodes (50, 80, and 100), computation time slightly increases as the number of robots increases. However, when there are 130 nodes, computation time decreases as robots increase from 3 to 5 but then gradually increases thereafter. Finally, for 150 nodes, computation time is highest when only 3 robots are used and slightly increases after 6 robots are utilized. These results demonstrate that as node count increases, solving TSP problems requires significantly more time. On the other hand, increasing robot count allows nodes to be allocated to individual robots via weighted Voronoi diagrams, reducing individual TSP problem sizes and providing computational advantages.

In summary, as discussed earlier, computational complexity is influenced by both weighted Voronoi diagrams and TSP problems in combination. As problem complexity

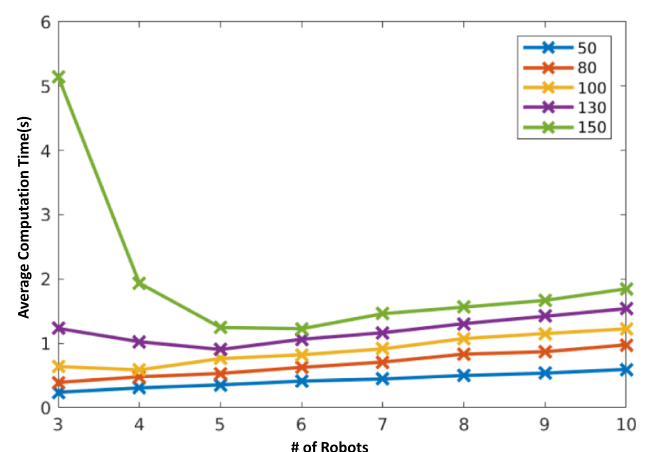


Fig. 11 Average computation time for each scenario. The x-axis represents the number of robots, and the y-axis shows the average computation time. Each line corresponds to a scenario with a different number of nodes required for area coverage. A subset of scenarios is displayed for clarity

increases, utilizing multiple robots helps moderate and alleviate issues related to increased computational load. However, depending on the scenario, there exists a point where computation time begins to increase with robot count. Thus, there may be limitations when using an excessively large number of robots. Furthermore, since this study employs an integer programming-based TSP solver, adopting other computationally efficient algorithms could further mitigate computation overhead caused by TSP problems more effectively.

7 Conclusion

In this study, we utilized a multi-robot system equipped with limited sensing capabilities to acquire information within a target area with no prior knowledge. The proposed algorithm aims to perform the search task with a multi-robot system, supplementing limited sensing capabilities through collaboration among the robots. To achieve this objective, we divided the path planning algorithm into two modules: the exploration module and the exploitation module. Each module was designed to address the characteristics required for the search task.

First, the FIM-based planner proposed in the author's previous study [25] was utilized as the exploitation path planner. Additionally, the uniform area distribution algorithm through the weighted Voronoi diagram with the weight adjusting method and the shortest distance coverage path planning based on TSP was proposed as the exploration path planner. Finally, a market-based selection algorithm was introduced to appropriately balance the trade-off between the exploration and exploitation paths.

To evaluate the performance of the proposed algorithm, Monte Carlo simulations were conducted. One hundred simulations were performed with random changes in the locations of landmarks and the starting position of the multi-robot system. In the simulations, we conducted a comparative analysis using the GA-based MTSP solution path generation algorithm and the lawnmower path generation algorithm as control groups. Through the simulations, it was confirmed that the proposed algorithm showed the best performance in terms of mission completion time and landmark location estimation accuracy.

The computational complexity of the proposed algorithm was analyzed through additional simulations conducted under varying conditions, including different robot counts and node densities. The results revealed that the complexity is influenced by both weighted Voronoi diagrams and TSP-based path planning. While increasing the number of robots reduces the size of individual TSP problems and alleviates computational overhead to some extent, this effect holds only

up to a certain point, beyond which additional robots may introduce coordination overhead and diminish computational benefits.

While these simulation results are promising, real-world implementation would introduce additional challenges related to robot control, inter-robot communication, and sensor data processing. A natural extension of this work would be to apply the proposed informative path planning framework to more complex real-world environments to assess its practical effectiveness. Such a transition from simulation to real-world application could help validate the approach and provide further insights for improving multi-robot search strategies in various fields, such as search and rescue operations, environmental monitoring, and autonomous exploration missions.

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Code or Data Availability Access to the data and code may be granted upon reasonable request and with permission from Korea Institute of Marine Science & Technology Promotion(KIMST).

Declarations

Ethics Approval Not applicable

Consent to Participate Not applicable

Consent for Publication Not applicable

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